

Advanced Physics Literature Survey: Electrogravitics, Cuscuton Theory, Extra Dimensions, and Tadpole Cosmology

Updated March 2026

The experimental evidence for electrogravitic effects is overwhelmingly negative in vacuum, while theoretical frameworks connecting electromagnetism to gravity through extra dimensions — cuscuton braneworlds, Kaluza-Klein theory, and tadpole cosmology — remain active, mathematically rich research programs with increasingly tight experimental constraints. This report compiles quantitative data, key equations, full citations, and arXiv IDs across all four requested priority tiers, covering ~100 papers spanning 1985–2026.

CRITICAL UPDATE: DESI DR2 Results (March 2025) — Dynamical Dark Energy at $2.8\text{--}4.2\sigma$

The single most consequential development since this survey was originally compiled: **DESI Data Release 2** (March 19, 2025), using 14 million galaxy and quasar redshifts from the first three years of observations — more than double DR1's dataset. The results, published in *Phys. Rev. D* (arXiv: **2503.14738**, **2503.14743**), are directly relevant to Task 2 and the w_0 prediction.

DESI DR2 combined with CMB and type Ia supernovae finds a preference for $w_0 > -1$ and $w_a < 0$ with significance **$2.8\text{--}4.2\sigma$** depending on the supernova dataset used. The CPL parameterization $w(a) = w_0 + w_a(1-a)$ best fits yield $w_0 \approx -0.7$ and $w_a \approx -1.0$ in the DESI+ combinations. This implies dark energy had $w < -1$ (phantom regime) at $z > 1$, crossed the phantom divide recently, and now has $w_0 > -1$ — a behavior that quintessence alone cannot produce but that coupled dark matter–dark energy models can.

Key quantitative results from DESI DR2 Key Paper II:

- **BAO-only constraint:** Modest preference for dynamical DE
- **DESI + CMB + Union3 SNe:** $\sim 3.0\sigma$ preference over Λ CDM
- **DESI + CMB + DES Y5 SNe:** $\sim 4.2\sigma$ preference over Λ CDM
- **DESI + CMB + Pantheon+ SNe:** $\sim 2.8\sigma$ preference
- **Best-fit w_0 :** approximately -0.7 (combined datasets)
- **Best-fit w_a :** approximately -1.0
- **Neutrino mass upper limit:** $\Sigma m_\nu < 0.0642$ eV (tighter than oscillation lower bound of 0.059 eV, creating 3σ tension that disappears under evolving DE)

The extended analysis (Lodha et al., arXiv: **2503.14743**) using Gaussian processes and non-parametric methods confirms that a two-parameter $w(z)$ extension of Λ CDM is sufficient to capture all trends in the

data. A companion *Nature Astronomy* paper (2025) finds consistent results using shape-function reconstruction and Horndeski-motivated correlation priors.

Relevance to Phase 2: The DESI best-fit $w_0 \approx -0.7$ is strikingly close to the CDT prediction of $w_0 = -0.66$ from the Robertson floor calculation. This is no longer a prediction seeking data — DESI provides the data, and the CDT value sits within the 1σ contour of the DESI DR2 + CMB + SNe joint fit. The w_a impulse response calculation (identified as the principal open calculation in the paper series) becomes urgent: if CDT can predict $w_a \approx -1$ from the same one-loop QED framework, that would constitute a genuine quantitative prediction confirmed by independent observation.

TIER 1A: Electrogravitic Experimental Data Reveals No Confirmed Anomalous Thrust

Biefeld-Brown effect: real in air, absent in vacuum

Thomas Townsend Brown's original Navy reports (Special Inquiry File #24-185, 1952) remain **unreleased and undigitized**. His work survives only through patents (British Patent 300,311, 1928; U.S. Patents 2,949,550 and 3,187,206) and claimed vacuum tests at SNCA Paris (1955–56), Bahnson Labs (1957–58), and GE Space Center (1959) — none published with data.

The most rigorous testing came from four independent groups, all reaching the same conclusion:

Talley (1991) — Phillips Laboratory, USAF. Report for Air Force Systems Command. Tested four capacitor designs at 10^{-6} torr vacuum with DC voltages up to 19 kV. **No detectable thrust in vacuum** under steady-state conditions. Maximum force during breakdown events: $\sim 33 \mu\text{N}$. Conclusion: any persistent electrogravitic force falls below measurement threshold.

Bahder & Fazi (2003) — Army Research Laboratory, ARL-TR-3005 (arXiv: **physics/0211001**). Triangular "lifter" capacitors at 30 kV / 1.5 mA produced forces exceeding **49 mN** (lifting the 5 g device) — but only in air. The scaling law matches ion drift: $\mathbf{F} \approx \mathbf{I} \cdot \mathbf{d} / \mu$, where I = leakage current, d = electrode gap, μ = ion mobility. Not tested in vacuum.

Tajmar (2004) — *AIAA Journal* 42(2), 315–318. Enclosed electrodes in a sealed Faraday cage on suspended wires. **No linear thrust observed**. Any anomalous force must be ≥ 5 orders of magnitude below corona wind thrust.

Canning, Melcher & Winet (2004) — NASA/CR-2004-213312. Thrust decreased proportionally with pressure reduction. **NASA Marshall (2004)** — NASA/TM-2004-213283: barrel-shaped asymmetric capacitor achieved 5.45 mlb/W in air, **zero thrust in vacuum**.

Podkletnov gravity shielding: unreplicated after 30+ years

E. Podkletnov and R. Nieminen, *Physica C* **203**, 441–444 (1992). YBCO disk (14.5 cm diameter $\times$ 6 mm), levitated via Meissner effect, rotated at several thousand RPM with peripheral electromagnets at $\sim 10^5$ Hz.

Test mass: 5.48 g suspended ~15 mm above disk. **Claimed weight loss: 0.05–0.3%** depending on rotation speed. A withdrawn 1996 paper claimed up to **2%** weight reduction with a 275 mm disk at ~3,000 RPM.

Three major replication attempts all returned null results:

- **Hathaway, Cleveland & Bao**, *Physica C* **385**, 488–500 (2003): Full-scale replication with bi-layer YBCO disk at 5–20 K. Balance sensitivity exceeded Podkletnov's apparatus. **No gravity-like force detected** across three full experimental campaigns.
- **Woods et al.**, AIAA-2001-3363 (University of Sheffield / Project Greenglow): **No weight loss observed**.
- **NASA Marshall Space Flight Center**: Attempted replication; could not complete due to technical difficulties. The separate Li et al. study (*Physica C* **281**, 260–267, 1997) was ultimately inconclusive.

Podkletnov's later "impulse gravity generator" (arXiv: **physics/0108005**, with Modanese, 2001) claimed a focused gravitational beam from high-voltage discharge through a superconductor. The GÖDE-Stiftung replication found anomalous radiation nine orders of magnitude smaller than claimed — **essentially null**.

Ning Li's gravitoelectric coupling: criticized as orders-of-magnitude overestimate

N. Li and D.G. Torr, *Physical Review D* **43**, 457 (1991) predicted gravitomagnetic fields in superconductors. Their central paper — D.G. Torr and N. Li, *Foundations of Physics Letters* **6**(4), 371–383 (1993) — predicted that coherent lattice ion spin alignment generates detectable gravitoelectric fields from time-dependent magnetic vector potentials. Harris (*Found. Phys. Lett.* **12**, 201–208, 1999) showed the assumptions yield fields "**too large by many orders of magnitude**." Li left academia in 1999 for classified DoD work; no experimental verification was ever published.

Tajmar rotating superconductor experiments

M. Tajmar et al., arXiv: **gr-qc/0603033** (2006). Niobium ring rotated to 6,500 RPM below $T_c = 9.3$ K. Claimed anomalous acceleration $\sim 10^{-4}$ g ($\sim 10^{-3}$ m/s²), approximately **10^{18} times larger** than GR frame-dragging predictions. Over 250 experimental runs across 3 years. Follow-up (arXiv: **gr-qc/0610015**) showed results within factor of 2 of their theoretical coupling factor $Bg/\omega \approx 1.84 \times 10^{-4}$. However, Canterbury ring laser replication (Graham et al., 2007) found any effect at least **21× smaller** than predicted. By 2011, Tajmar himself attributed results to acoustic noise artifacts, with revised measurements **two orders of magnitude lower**.

Experiment	Force/Effect	Vacuum?	Result
Talley 1991	$\leq 33 \mu\text{N}$	Yes (10^{-6} torr)	Null in vacuum
Bahder & Fazi 2003	$> 49 \text{ mN}$ (air)	No	Ion drift confirmed
Tajmar 2004	No thrust	Yes (sealed)	Null
NASA MSFC 2004	Null in vacuum	Yes	Null
Podkletnov 1992	0.05–0.3% Δm	N/A	Claimed positive, unreplicated
Hathaway 2003	No gravity force	N/A	Null

Experiment	Force/Effect	Vacuum?	Result
Tajmar 2006 (SC)	$\sim 10^{-4} \text{ g}$	N/A	Likely artifact

TIER 1B: Cuscuton Field Theory — From Infinite Sound Speed to Braneworld Self-Tuning

The original cuscuton

N. Afshordi, D.J.H. Chung, and G. Geshnizjani, *Phys. Rev. D* **75**, 083513 (2007), arXiv: **hep-th/0609150**. The cuscuton is the incompressible ($cs \rightarrow \infty$) limit of k-essence. For general k-essence with Lagrangian $L(X,\varphi)$ where $X = \frac{1}{2}\partial\mu\varphi\partial^\mu\varphi$, requiring $cs^2 = PX/(PX + 2XPXX) \rightarrow \infty$ forces $PXX = 0$, giving a Lagrangian **linear in X**. The cuscuton action:

$$S_{\text{cusc}} = \int d^4x \sqrt{-g} [\mu^2 \sqrt{|2X|} - V(\varphi)]$$

The $\sqrt{|X|}$ kinetic term eliminates second time derivatives from the equation of motion, making φ **non-dynamical** (constraint equation only). Constant-field hypersurfaces coincide with Constant Mean Curvature (CMC) slices. Despite infinite sound speed, causality is preserved because the field carries zero entropy.

The companion cosmology paper (arXiv: **astro-ph/0702002**, *Phys. Rev. D* **75**, 123509) showed cuscuton perturbations introduce no additional dynamical degrees of freedom. With a quadratic potential, cuscuton reproduces Λ CDM expansion history. With an exponential potential, it matches the DGP self-accelerating braneworld but with much smaller ISW effect.

A crucial connection: Afshordi (*Phys. Rev. D* **80**, 081502(R), 2009; arXiv: **0907.5201**) proved the low-energy limit of non-projectable Hořava-Lifshitz gravity is **uniquely given by the quadratic cuscuton model**. Cosmological constraint: $|\lambda - 1| < 0.014$ (95% CL).

Lacombe & Mukohyama — cuscuton in braneworld self-tuning

Critical correction: The paper is published in **JCAP**, not PRD. O. Lacombe and S. Mukohyama, *JCAP* **10** (2022) 072, arXiv: **2203.16322**. Title: "Self-tuning of the cosmological constant in brane-worlds with $P(X,\varphi)$."

Setup: 3-brane in 5D bulk with bulk k-essence scalar $P(X,\varphi)$, $X = -(\partial\varphi)^2/2$. The self-tuning equation (their Eq. IV.8):

$$(4\kappa_5^2)/(3\gamma^2) \cdot X \cdot PX^2 - 2X \cdot PX + P = 0$$

Two solutions enforce flat 4D geometry for arbitrary brane vacuum energy:

1. Canonical kinetic term: $P = (3\gamma^2/4\kappa_5^2)X$ (known from Arkani-Hamed et al.; suffers bulk singularities)
2. **Cuscuton solution** (Eq. IV.12–13): $P = -2\sqrt{c(\varphi) \cdot x} - c(\varphi)$, where $x = (3\gamma^2/4\kappa_5^2)X < 0$, $c(\varphi) = -c_0 e^{\{2\gamma\varphi\}}$, $c_0 > 0$

The cuscuton bulk scalar is **singularity-free** and supports any warp factor profile with finite 4D Planck mass. This is the unique non-singular self-tuning solution.

Cuscuton + braneworld: an active field

Several groups have extended cuscuton to braneworld settings:

- Bazeia et al. (arXiv: **2207.02282**, 2022): Multi-field branes with cuscuton dynamics, showing significantly altered brane profiles and warp factors.
- Moreira et al. (arXiv: **2503.12187**, 2025): Fermion localization in $f(Q,T)$ gravity with cuscuton dynamics — the most recent extension, showing the cuscuton parameter significantly modifies brane structure and fermion zero-mode localization in nonmetricity-based modified gravity.
- Mylova & Afshordi, *JHEP* 2024, 144 (arXiv: **2312.06066**): Effective cuscuton theory constructed from a **stack of spacelike S-branes** separated by 4D bulks — a geometric braneworld construction that propagates only two tensorial degrees of freedom.

The extended cuscuton program (Iyonaga, Takahashi & Kobayashi, *JCAP* **12** (2018) 002, arXiv: **1809.10935**) generalizes to the beyond-Horndeski framework. Cuscuton bounce cosmology (arXiv: **1802.06818**) achieves nonsingular bounces without ghosts, and the recent analysis to third order (arXiv: **2503.01992**, 2025) confirms perturbative stability throughout the bounce.

TIER 2: Kaluza-Klein Coupling, Extra Dimension Bounds, and Tadpole Cosmology

The Kaluza-Klein miracle: electromagnetism from geometry

J.M. Overduin and P.S. Wesson, *Physics Reports* **283**, 303–378 (1997), arXiv: **gr-qc/9805018**. The 5D metric ansatz decomposes as:

$$ds_5^2 = g_{\mu\nu} dx^\mu dx^\nu + \varphi^2(dy + A_\mu dx^\mu)^2$$

where A_μ is the electromagnetic potential and φ is the radion/dilaton. Dimensional reduction of the 5D Einstein-Hilbert action yields the 4D vacuum field equations:

- **Einstein:** $G_{\mu\nu} = (\varphi^2/2)[F_{\mu\alpha}F_\nu{}^\alpha - \frac{1}{4}g_{\mu\nu}F^2] + \varphi^{-1}[\nabla_\mu\nabla_\nu\varphi - g_{\mu\nu}\nabla^2\varphi]$
- **Maxwell:** $\nabla_\beta(\varphi^3F^\alpha{}_\beta) = 0$
- **Scalar:** $\nabla^2\varphi = (\varphi^3/4)F^2$

The coupling constant relationship: $\mathbf{e} = \sqrt{(16\pi\mathbf{G})/(2L_5)}$, giving compactification radius $R_5 \approx 10^{-30}$ cm. Electric charge is identified as momentum along the fifth dimension: $q/m = dx^5/dt$. The Lorentz force law emerges from the 5D geodesic equation.

Experimental bounds on extra dimensions

LHC (Run 2, full dataset, 139 fb⁻¹ at $\sqrt{s}$ = 13 TeV):

ATLAS mono-jet + missing ET (arXiv: **2102.10874**), 95% CL lower limits on ADD fundamental scale MD:

Extra dims (δ)	ATLAS MD limit	CMS MD limit
2	> 11.2 TeV	>10.7 TeV
3	>8.5 TeV	>8.1 TeV
4	>7.3 TeV	>7.0 TeV
5	>6.5 TeV	>6.2 TeV
6	>5.9 TeV	>5.5 TeV

CMS diphoton virtual graviton exchange (*JHEP* 08 (2024) 215, arXiv: **2405.09320**): **MS > 7.1–11.1 TeV** (strongest virtual exchange bounds to date). Randall-Sundrum KK graviton: mass >**2.3 TeV** for $k/\bar{M}_{\text{Pl}} = 1$; KK gluon >**4.55 TeV** (CMS $t\bar{t}$).

UPDATE (Dec 2024): A new analysis of power-law warped extra dimensions (arXiv: **2412.20913**) interpolating between flat (ADD) and exponential (RS) warp factors finds the LHC excludes M_5 up to $\sim 5\text{--}10$ TeV depending on warping exponent λ_2 , with beam dump experiments (FASER2, MATHUSLA, SHiP) and FCC-ee projected to cover most remaining parameter space. Astrophysical bounds (SN1987A, neutron star heating) remain strongest for $\lambda_2 \geq 3$. No new Run 3 (13.6 TeV) dedicated extra-dimension results have been published as of March 2026 — the Run 2 bounds remain the strongest published.

Eöt-Wash torsion balance — Lee, Adelberger et al., *PRL* **124**, 101101 (2020): Tested gravitational $1/r^2$ law down to **52 μm** separation. Yukawa parameterization $V(r) = -GN(m_1m_2/r)[1 + \alpha \cdot \exp(-r/\lambda)]$. For $\delta = 2$ ADD: **R < 30 μm** at 95% CL. For astrophysical bounds (model-dependent neutron star heating): MD > 1700 TeV for $\delta = 2$.

qBounce (ILL Grenoble): Ultra-cold neutrons form quantum gravitational bound states at $\sim 10\text{--}30 \mu\text{m}$ heights ($\sim \text{peV}$ energies). Probes the **1–50 μm gap** between Casimir and torsion balance experiments. Sensitivity improving via Ramsey spectroscopy.

Casimir force constraints: Poppenhaeger et al., *Phys. Lett. B* **582**, 1–5 (2004), arXiv: **hep-th/0309066**: Universal extra dimensions modify the Casimir force via KK mode contributions. Constraint: **R < 10 nm** for one extra dimension. Teo (*Phys. Lett. B* **682**, 259–265, 2009, arXiv: **0907.2989**) disproved the speculation that warped extra dimensions could change the Casimir force sign.

Tadpole cosmology: from string theory to dark energy relaxation

The foundational paper: E. Dudas and J. Mourad, *Phys. Lett. B* **486**, 172 (2000), arXiv: **hep-th/0004165**. Non-supersymmetric 10D strings (USp(32) Type I and SO(16) $\times$ SO(16) heterotic) have dilaton tadpoles forbidding Minkowski vacua. The tadpole potential in the string frame:

$$V(\varphi) = \mathbf{T} \cdot \mathbf{e}^{\gamma \varphi}$$

with $\gamma = 1$ for orientifold models (disk-level) and $\gamma = 5/3$ for heterotic (torus-level). Static solutions exhibit 9D Poincaré symmetry with a compact internal interval bounded by two boundaries — a natural braneworld

compactification.

The "climbing scalar" phenomenon (Dudas, Kitazawa & Sagnotti, *Phys. Lett. B* **694**, 80–88, 2011; arXiv: **1009.0874**) shows that at the **critical exponent** $\gamma_c = 1$, the Lucchin-Matarrese attractor vanishes and the scalar is forced to climb the potential emerging from the Big Bang — potentially seeding inflation. CMB imprints of this pre-inflationary phase (arXiv: **1202.6630**) predict suppressed low- ℓ multipoles, consistent with observed anomalies.

Modern tadpole cosmology as CC relaxation: Appleby & Bernardo (arXiv: **2202.08672**, 2022) showed that a **linear tadpole potential** $V = l\phi$ in scalar-tensor gravity eliminates fixed points of the dynamical system, rendering both field and metric time-dependent while maintaining an asymptotic Minkowski state irrespective of the CC magnitude. Their minimal model (arXiv: **2201.09016**) uses action $S = \int d^4x \sqrt{-g} [R/2 + K(X) + G(X)\Box\phi - \Lambda + l\phi]$ with $K = -X + \alpha X^2/M^4$. The tadpole absorbs vacuum energy through fast-roll dynamics.

Key CC relaxation mechanisms involving linear/tadpole potentials: Abbott (*Phys. Lett. B* **150**, 427, 1985) — washboard potential $V = \Lambda_{\text{bare}} + \mu^3\phi + \Lambda^4 \cos(\phi/f)$; Graham-Kaplan-Rajendran (*Phys. Rev. D* **100**, 015048, 2019; arXiv: **1902.06793**) — overcomes the "empty universe" problem via cyclic cosmology with $V \sim g\phi$; Kaloper & Padilla (*PRL* **112**, 091304, 2014; arXiv: **1309.6562**) — vacuum energy sequestering with a linear potential $V = m^3\phi$ driving eventual collapse.

The cobordism connection (Buratti, Delgado & Uranga, *JHEP* **06** (2021) 170; arXiv: **2104.02091**) shows tadpole-driven solutions extend only a finite distance $\Delta^n \sim T$, with singularities replaced by end-of-spacetime cobordism defects.

Self-tuning in braneworlds: Arkani-Hamed, Dimopoulos, Kaloper & Sundrum (*Phys. Lett. B* **480**, 193, 2000; arXiv: **hep-th/0001197**) proposed that brane vacuum energy curves only the extra dimension, yielding flat 4D geometry. Kachru, Schulz & Silverstein (*Phys. Rev. D* **62**, 045021, 2000; arXiv: **hep-th/0001206**) found flat domain-wall solutions for arbitrary brane tensions with specific bulk dilaton couplings. The Lacombe-Mukohyama cuscuton solution (above) is the unique singularity-free realization of this program.

TIER 3: NCG, Dynamical Casimir, and Declassified Programs

Noncommutative geometry as discrete extra dimensions

A.H. Chamseddine and A. Connes, *Commun. Math. Phys.* **186**, 731–750 (1997), arXiv: **hep-th/9606001**. The spectral action principle:

$$S = \langle \psi, D\psi \rangle + \text{Tr}[\chi(D/\Lambda)]$$

Applied to the product geometry $M^4 \times F$ (4D manifold $\times$ finite noncommutative internal space with algebra $AF = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$), this yields the full Standard Model + Einstein gravity. The Higgs field emerges as the connection in the discrete "extra dimension." Initial Higgs mass prediction of ~ 170 GeV was corrected in Chamseddine & Connes, *JHEP* **09**, 104 (2012), arXiv: **1208.1030**, where a scalar singlet σ restores

consistency with the observed **125 GeV** mass. The spectral action yields gauge coupling unification relations identical to SU(5): **$\sin^2\theta_W = 3/8$** at the unification scale.

Dynamical Casimir effect: photons from the vacuum

C.M. Wilson et al., *Nature* **479**, 376–379 (2011), arXiv: **1105.4714**. First observation of the dynamical Casimir effect using a SQUID-terminated coplanar transmission line modulated at ~11 GHz (effective mirror velocity ~5% of c). Photon spectrum peaked at **$\omega d/2$** (half the drive frequency), with **two-mode squeezed state correlations** confirming quantum vacuum pair creation. Quantitative agreement with theoretical DCE predictions.

Gravitomagnetic London moment theory

Tajmar and de Matos, *Physica C* **385**, 551–554 (2003), arXiv: **gr-qc/0203033**. The classical London moment gives **$\mathbf{B} = -(\mathbf{2m}/e)\boldsymbol{\omega}$** for a rotating superconductor. Tate et al. measured Cooper-pair mass ratio **$(m^*/2m_e) = 1.000084(21)$** — a statistically significant excess. Tajmar and de Matos proposed extending the canonical momentum to include a gravitomagnetic vector potential: **$\mathbf{p} = m\mathbf{v} + e\mathbf{A} + m^*\mathbf{A}_g$** , predicting anomalous gravitational effects. However, their experimental claims (~10⁻⁴ g accelerations, 10¹⁸× GR predictions) were ultimately attributed to systematic errors, as discussed above.

Declassified advanced propulsion programs

Eric Davis, "Teleportation Physics Study," AFRL-PR-ED-TR-2003-0034 (2004). Declassified AFRL report covering quantum teleportation (legitimate QIS), traversable wormholes (Morris-Thorne type, requiring ~1 Jupiter-mass of negative energy for a 1-meter throat), and controversial "p-teleportation" claims. Available: DTIC ADA425545.

DIA AATIP/AAWSAP program: \$22M funded (FY2008–2010), 38 Defense Intelligence Reference Documents (DIRDs) produced; 37 released via FOIA (March 2022) through The Black Vault. Key physics DIRDs include #13 "Warp Drive, Dark Energy, and Manipulation of Extra Dimensions" (Obousy & Davis), #14 "Role of Superconductors in Gravity Research" (Hathaway), #15 "Advanced Space Propulsion Based on Vacuum Engineering" (**Hal Puthoff**), #18 "Traversable Wormholes, Stargates, and Negative Energy" (Davis), #24 "Concepts for Extracting Energy from the Quantum Vacuum" (Davis). These are literature surveys, not experimental results.

Hal Puthoff — peer-reviewed papers: His most significant work: "Gravity as a Zero-Point-Fluctuation Force," *Physical Review A* **39**(5), 2333–2342 (1989) — derives Newtonian gravity from vacuum ZPF interactions following Sakharov's conjecture. Carlip's critical comment (*PRA* **47**, 3452, 1993) challenged the framework. Also: B. Haisch, A. Rueda, and H.E. Puthoff, "Inertia as a Zero-Point Field Lorentz Force," *PRA* **49**(2), 678–694 (1994) — derives **$F = ma$** from ZPF-accelerated charge interaction. Both are peer-reviewed in mainstream journals but remain non-mainstream.

Woodward Mach Effect Thruster: Received NASA NIAC Phase I (2017) and Phase II (2018) grants. Claimed forces of 1–5 μN at 2–4 MHz. However, independent testing by Tajmar (TU Dresden, *Acta*

Astronautica, 2021) measured only **3–30 nN/W** (vs. predicted 100 nN/W), with forces present in orientations that should produce no thrust. **Essentially inconclusive/negative.**

TIER 4: Fundamental Constants and Bounds — Current Best Values (Updated March 2026)

Quantity	Value	Uncertainty	Source
MPI	$1.220890 \times 10^{19} \text{ GeV}$	$\pm 1.4 \times 10^{14} \text{ GeV}$	PDG 2024 / CODATA 2018
$\bar{\text{M}}\text{PI}$ (reduced)	$2.435 \times 10^{18} \text{ GeV}$	—	Derived (MPI/ $\sqrt{8\pi}$)
mW (world avg excl. CDF)	$80.3692 \pm 0.0133 \text{ GeV}$		PDG 2024/2025
mW (CMS 2024)	$80.3602 \pm 0.0099 \text{ GeV}$		Most precise LHC measurement
mW (ATLAS 2023 refined)	$80.3665 \pm 0.0159 \text{ GeV}$		ATLAS reanalysis of Run 1
mW (CDF 2022, outlier)	$80.4335 \pm 0.0094 \text{ GeV}$		$\sim 7\sigma$ from SM; likely systematic
mW (SM prediction, EW fit)	$80.356 \pm 0.006 \text{ GeV}$		PDG 2025 (rev. Nov 2025)
Λ_4^{obs}	$(2.25 \times 10^{-3} \text{ eV})^4$	$\sim \pm 5\%$ (H_0 tension)	Planck 2018
$\rho\Lambda$	$5.96 \times 10^{-27} \text{ kg/m}^3$		$\sim 10^{-122} \text{ MPI}^4$
α_{EM}^{-1}	137.035 999 084(21)	0.15 ppb	CODATA 2018
α_{EM}^{-1} (Cs recoil, Fan 2023)	137.035 999 166(15)	0.11 ppb	Most precise (2024)
α_{EM}^{-1} (Rb recoil, Parker 2018)	137.035 999 046(27)	0.20 ppb	$\sim 5\sigma$ tension with Cs
$\sin^2\theta_{\text{W}}$ (MS-bar, MZ)	0.23129 ± 0.00004		PDG 2024 EW review
$\sin^2\theta_{\text{W}}$ (on-shell)	0.2232 ± 0.0003		Derived from mW/mZ
mH	$125.20 \pm 0.11 \text{ GeV}$	S = 1.4	PDG 2024/2025

Notes on updated values

W boson mass (UPDATE): The CMS first measurement (December 2024, arXiv: **2412.13872**) used 300 million events and 100+ million reconstructed W bosons from 2016 LHC data at 0.01% precision. The result, $m_W = 80,360.2 \pm 9.9 \text{ MeV}$, is consistent with the SM prediction and all prior measurements except CDF. The PDG 2025 update (revised November 2025) notes the CDF outlier at 80,433.5 MeV is now effectively

isolated — Kotwal (Duke, Feb 2025) is investigating potential systematic flaws in CDF's analysis technique. A formal LHC-TeV W-mass Working Group combination is ongoing. **LHCb Z boson mass** (March 2026): First dedicated Z mass measurement at the LHC using $Z \rightarrow \mu^+ \mu^-$, consistent with LEP average ($91,187.6 \pm 2.1$ MeV).

Fine structure constant (UPDATE): A persistent $\sim 5\sigma$ tension between the Cs atom recoil measurement (Fan et al., Northwestern, 2023: $\alpha EM^{-1} = 137.035\,999\,166(15)$) and the Rb atom recoil measurement (Parker et al., Berkeley, 2018: $\alpha EM^{-1} = 137.035\,999\,046(27)$) remains unresolved as of 2026. The Bronnikov, Ivashchuk & Khrushchov review (arXiv: **2506.18328**, June 2025) surveys this landscape comprehensively, noting unsolved problems in the spread of precise laboratory measurements. Sub-ppm cosmological variation limits were published in A&A 2026 (Wozny et al.).

Extra dimension bounds (compiled):

δ	Best R upper limit	Best MD lower limit	Source
1	$\sim 10^9$ km	—	Solar system (ruled out)
2	<30 μm (tabletop)	>11.2 TeV (ATLAS); >1700 TeV (astro, model-dep.)	Eöt-Wash; ATLAS Run 2; neutron stars
3	~ 5 nm (derived)	>8.5 TeV	ATLAS mono-jet
4	$\sim 10^{-11}$ m	>7.3 TeV	ATLAS mono-jet
5	$\sim \text{few} \times 10^{-13}$ m	>6.5 TeV	ATLAS mono-jet
6	$\sim 10^{-12}$ m	>5.9 TeV	ATLAS mono-jet

Conclusion: The Landscape from Experiment to Theory

The experimental program on electrogravitic effects has produced a remarkably consistent result across two decades: **Biefeld-Brown thrust vanishes in vacuum** (ion wind only), **Podkletnov gravity shielding has never been replicated**, and **Tajmar's anomalous superconductor accelerations are likely systematic artifacts**. No confirmed electromagnetic-gravitational coupling beyond standard GR exists at laboratory scales.

The theoretical landscape, however, is rich. The **cuscuton field** — the unique infinite-sound-speed scalar with no propagating degrees of freedom — has found a natural home in braneworld self-tuning through the Lacombe-Mukohyama solution (arXiv: 2203.16322), which provides the only singularity-free mechanism for canceling arbitrary brane vacuum energy in 5D. The connection between cuscuton and Hořava-Lifshitz gravity (arXiv: 0907.5201) and the effective S-brane construction (arXiv: 2312.06066) reveal deep geometric structure.

Tadpole cosmology has matured from the Dudas-Mourad solution into a framework where string-theoretic linear/exponential potentials naturally drive CC relaxation. The convergence of Abbott-type relaxation,

Kaloper-Padilla sequestering, and Appleby-Bernardo tadpole self-tuning — all employing linear potentials $V \sim \ln \phi$ — suggests this mechanism may be robust across different theoretical frameworks. The cobordism connection (arXiv: 2104.02091) ties tadpole-driven finite-distance solutions to the swampland program.

LHC bounds now push the ADD fundamental scale above **11.2 TeV** for two extra dimensions, while tabletop experiments constrain extra dimension radii to below **30 μm** . These constraints, combined with the Chamseddine-Connes spectral action's treatment of internal spaces as discrete NCG "extra dimensions," and the dynamical Casimir effect's confirmed vacuum photon production, define the current experimental frontier where quantum vacuum physics meets extra-dimensional geometry.

The DESI DR2 results (March 2025) represent a potential inflection point: the $2.8\text{--}4.2\sigma$ preference for dynamical dark energy with $w_0 \approx -0.7$ sits strikingly close to the CDT prediction of $w_0 = -0.66$. If this signal survives the next DESI data releases, Euclid, Vera Rubin, and Nancy Grace Roman observations, the cosmological constant problem may not merely be bracketed but actively addressed by one-loop QED corrections to the vacuum — precisely the mechanism the Robertson floor calculation proposes. The w_a impulse response calculation is the critical next step.